

Land Degradation Classification

Model Comparison Report — Uttar Pradesh (2020–2024)

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Baseline Model Comparison

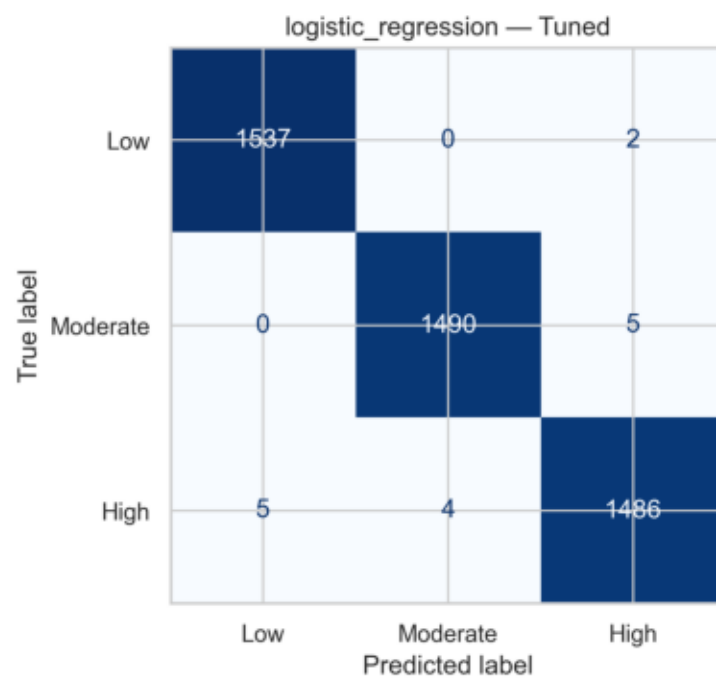
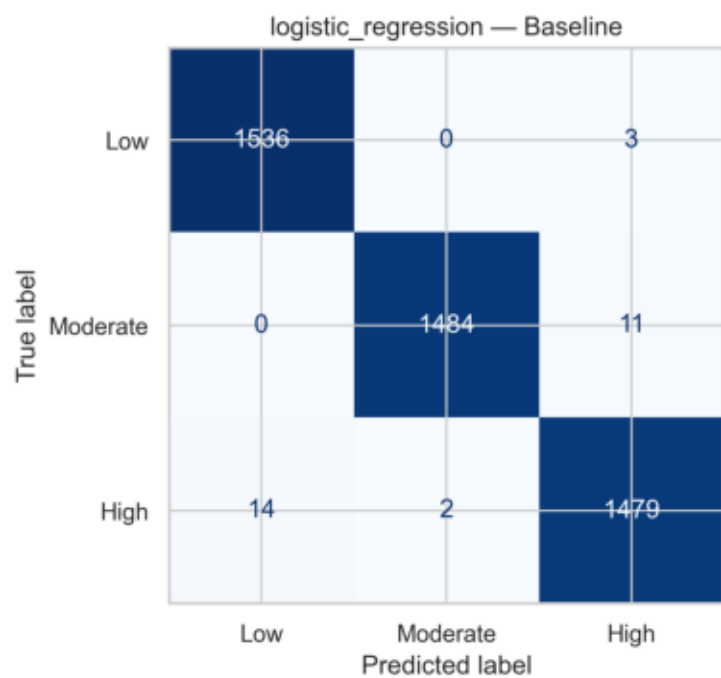
rank	model	accuracy	precision	recall	f1_score	roc_auc	train_time_sec	predict_time_sec
1	logistic_regression	0.9934	0.9934	0.9933	0.9934	0.9999	0.5876	0.0085
2	xgboost	0.9709	0.9708	0.9707	0.9708	0.9982	1.9125	0.0251
3	random_forest	0.9574	0.9577	0.9572	0.9574	0.9957	5.5802	0.3591
4	gradient_boosting	0.9455	0.9463	0.9454	0.9456	0.9947	44.0007	0.0421
5	decision_tree	0.9221	0.9221	0.9218	0.9219	0.9414	0.8897	0.0047

Baseline vs Tuned — Test Set

model	variant	accuracy	f1_score	roc_auc
logistic_regression	baseline	0.9934	0.9934	0.9999
logistic_regression	tuned	0.9965	0.9965	1.0
xgboost	baseline	0.9709	0.9708	0.9982
xgboost	tuned	0.9735	0.9734	0.9985

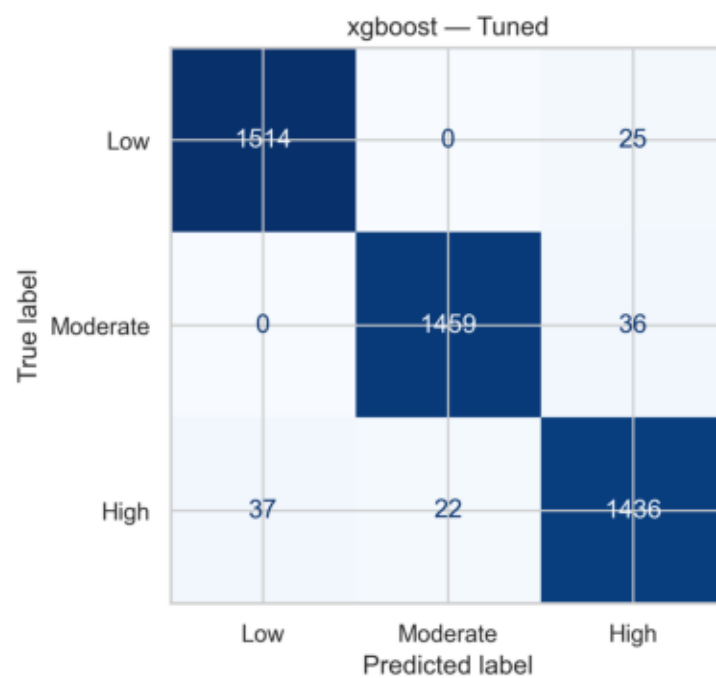
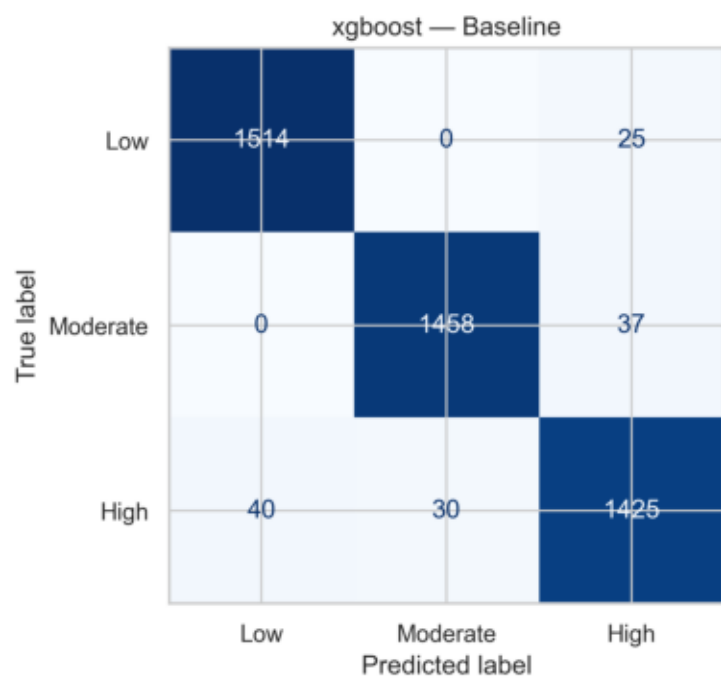
Confusion Logistic Regression

Confusion Matrices: logistic_regression

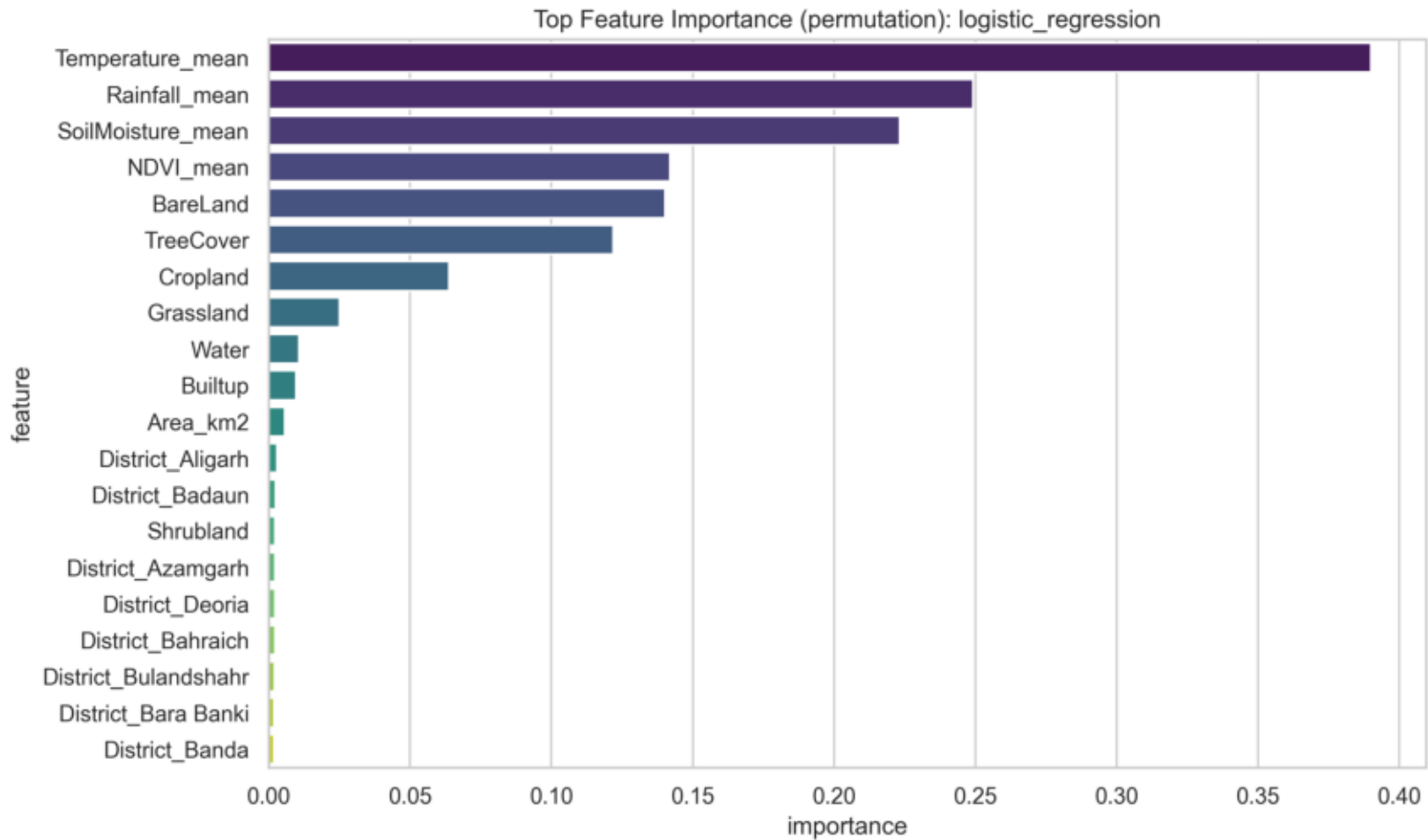


Confusion Xgboost

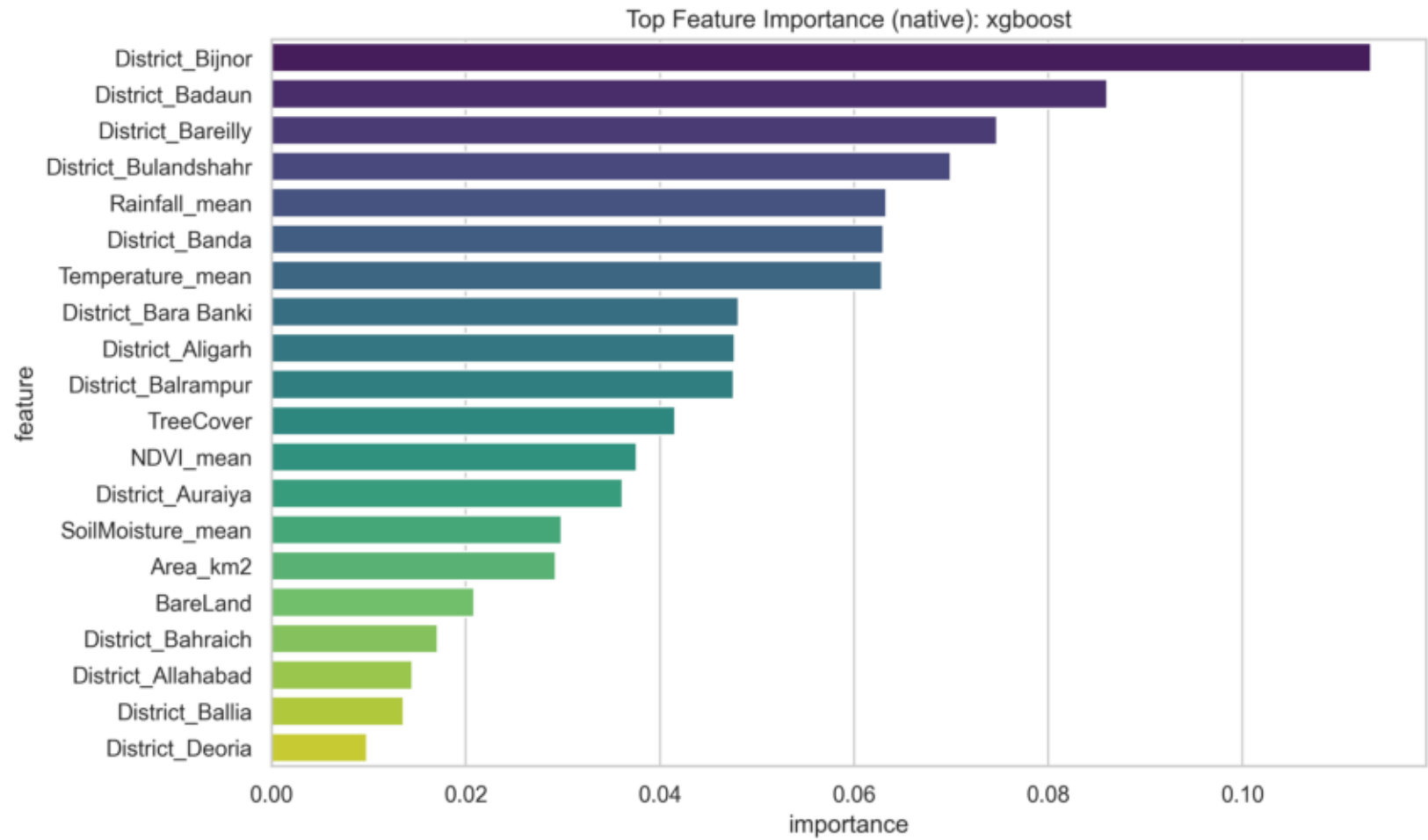
Confusion Matrices: xgboost



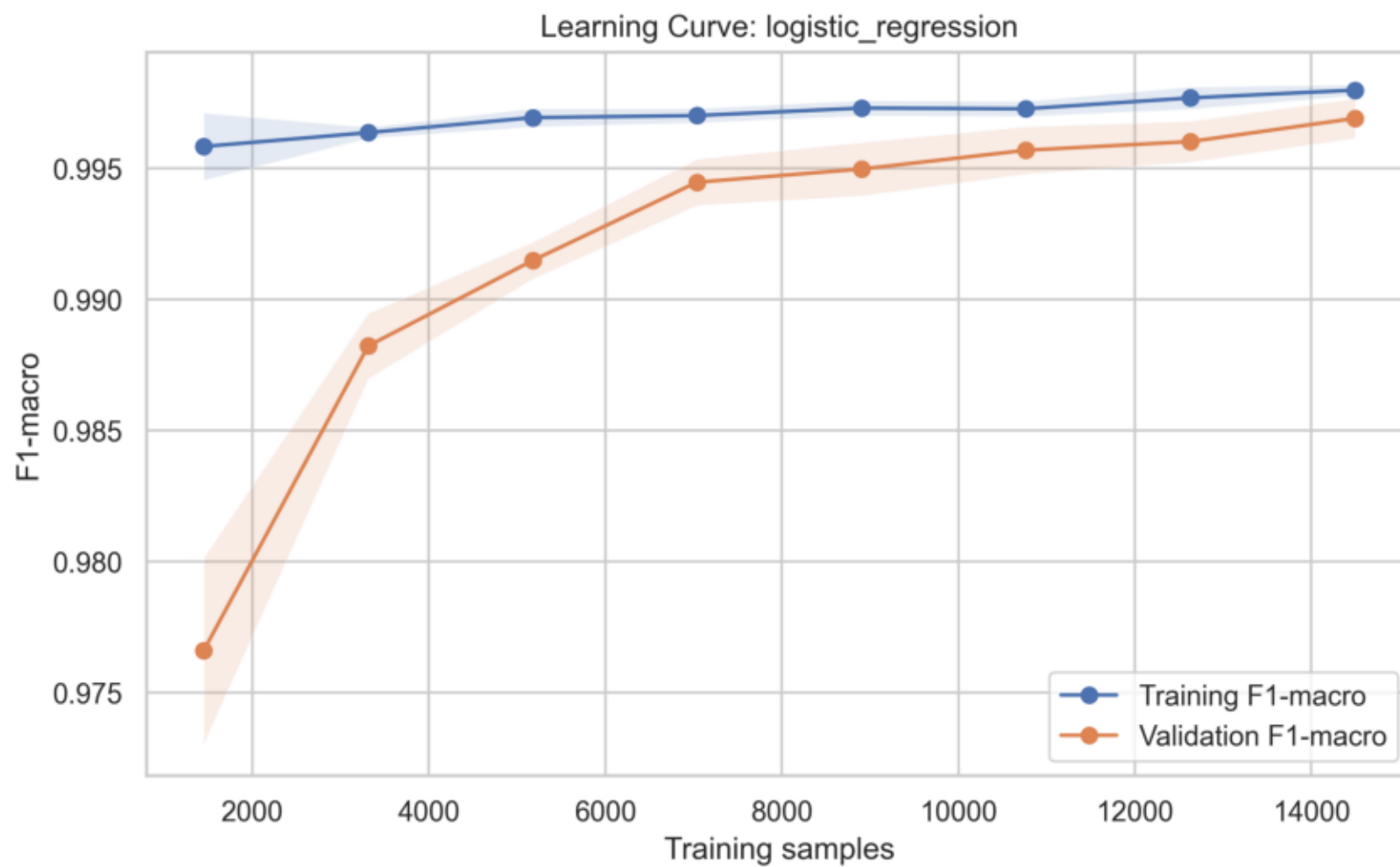
Feature Importance Logistic Regression



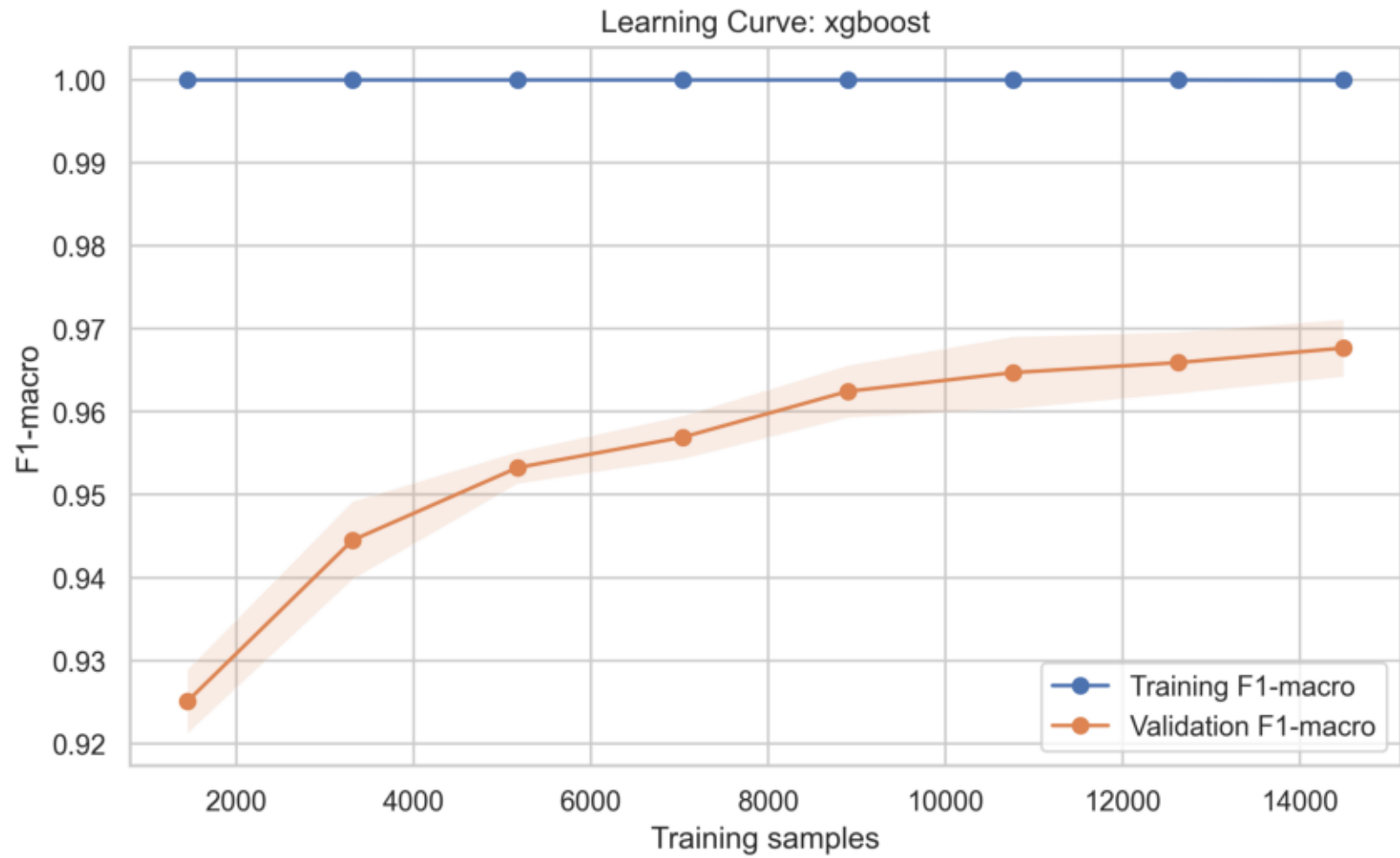
Feature Importance Xgboost



Learning Curve Logistic Regression

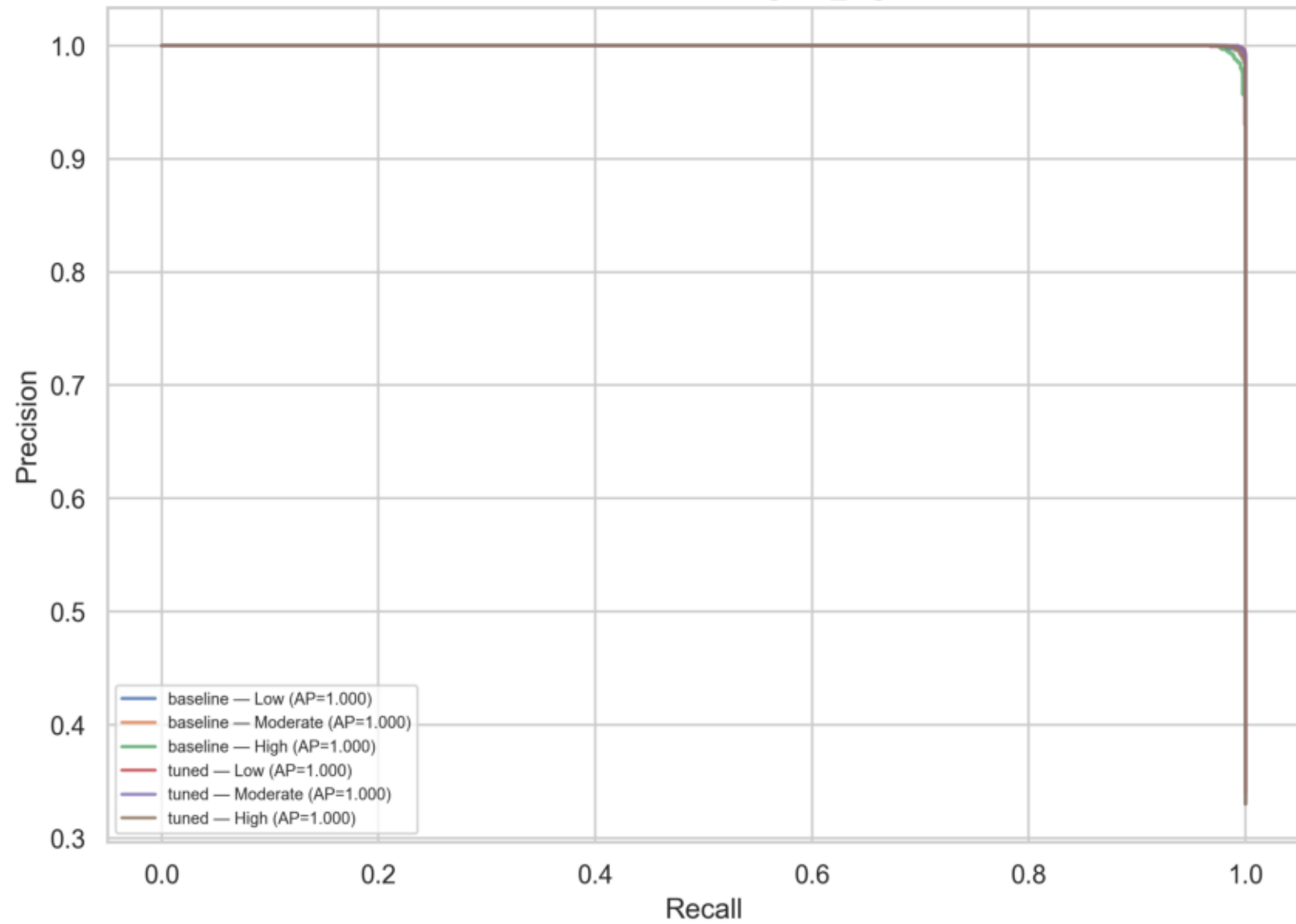


Learning Curve Xgboost



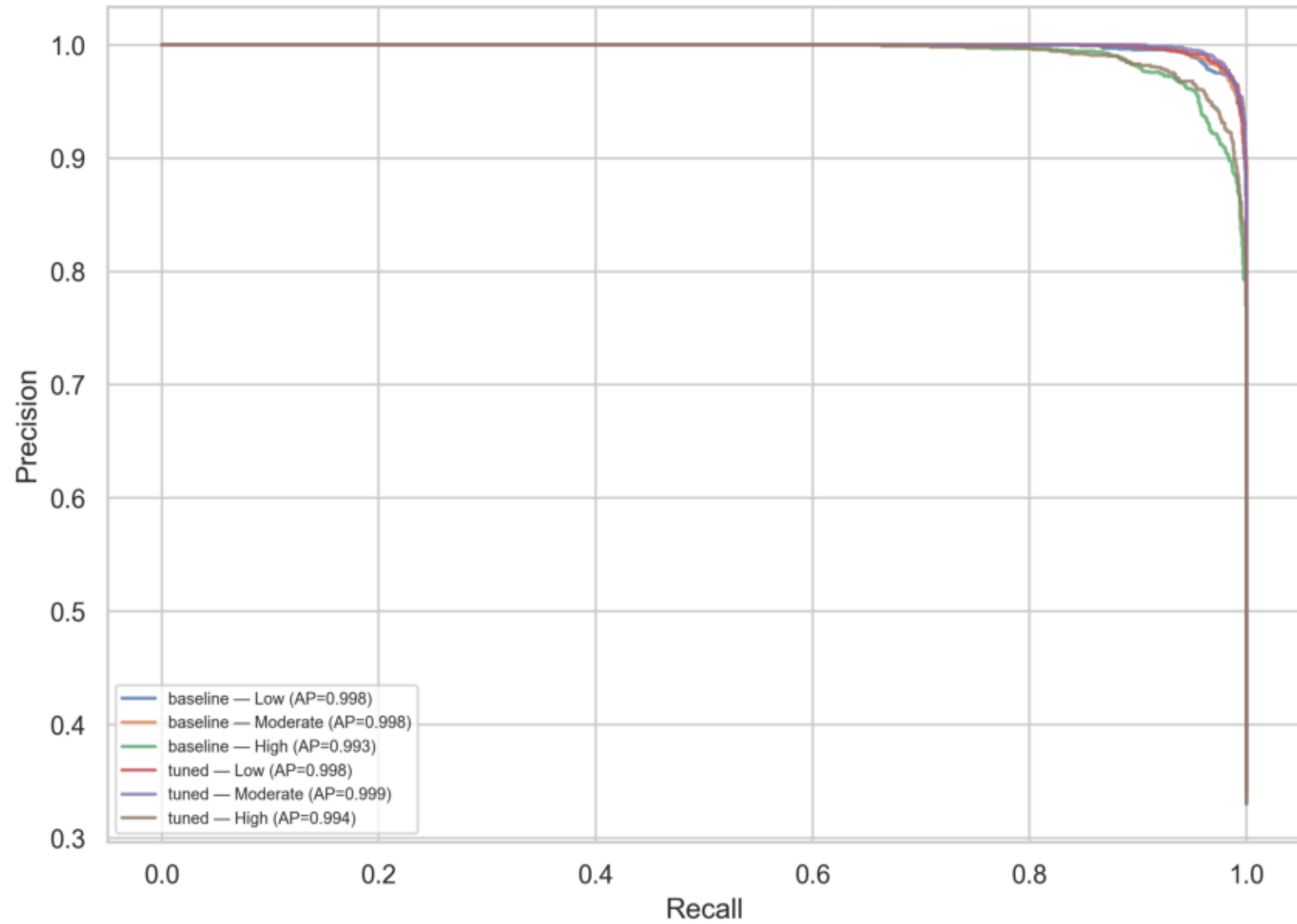
Pr Logistic Regression

Precision-Recall Curves: logistic_regression



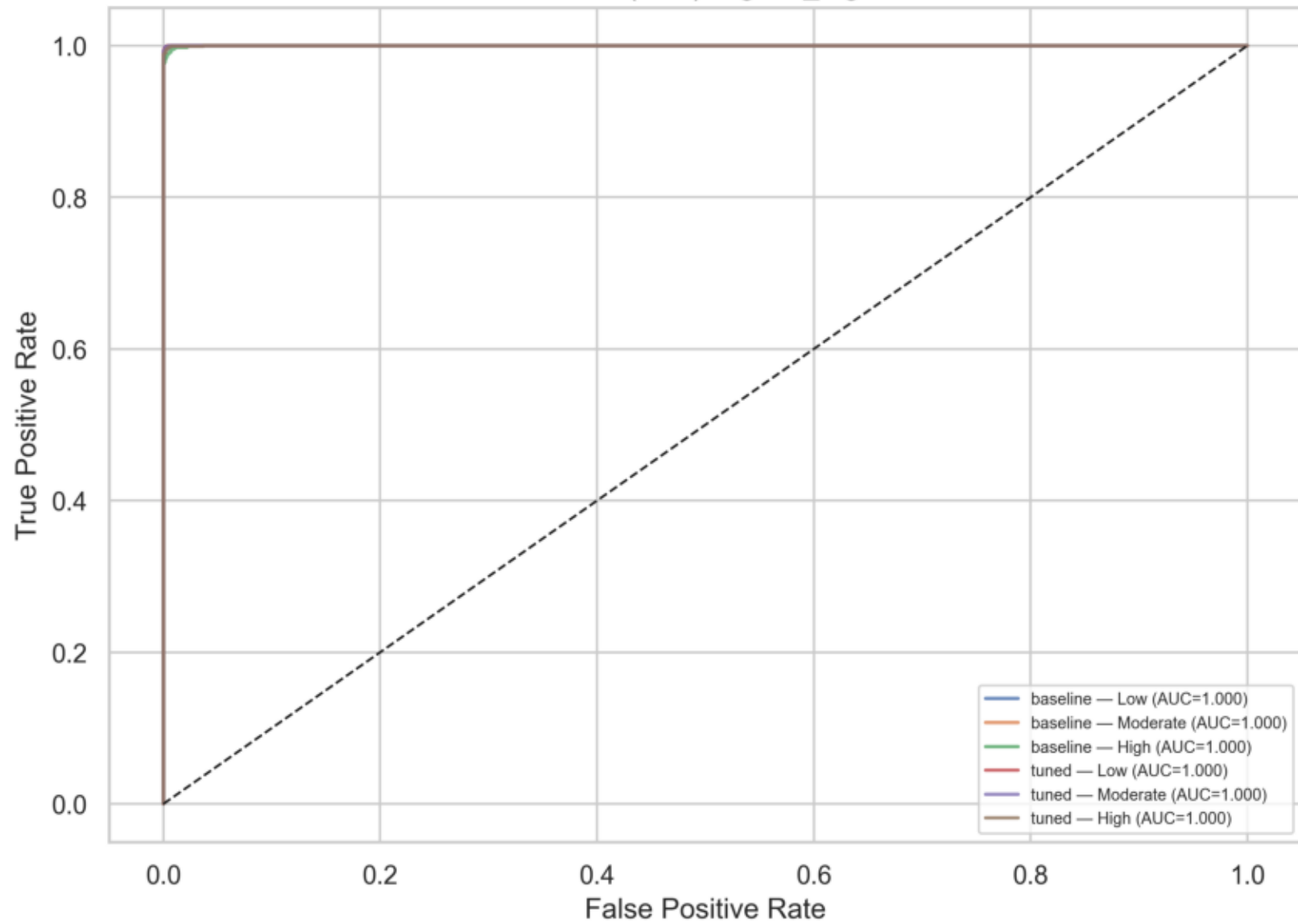
Pr Xgboost

Precision-Recall Curves: xgboost



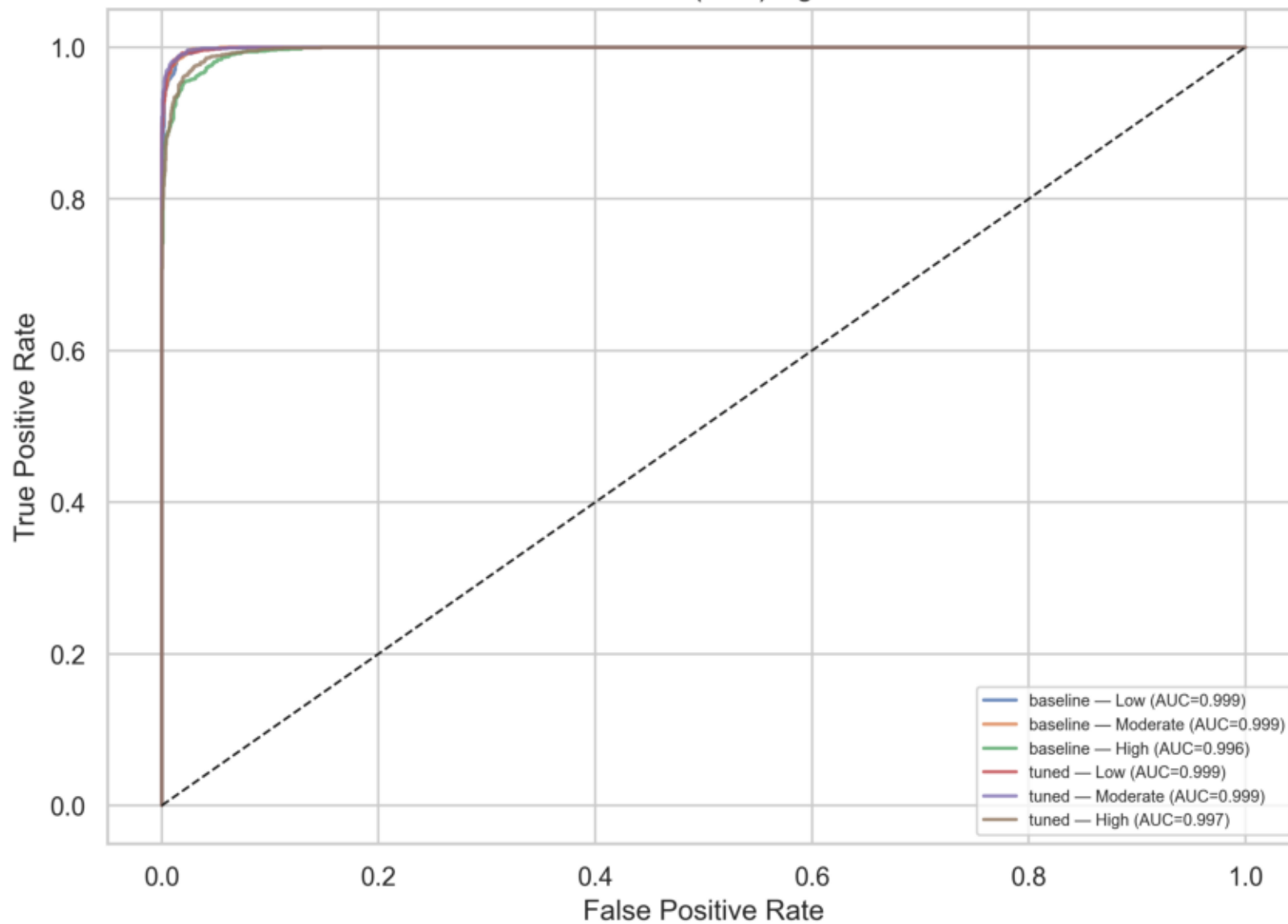
Roc Logistic Regression

ROC Curves (OvR): logistic_regression



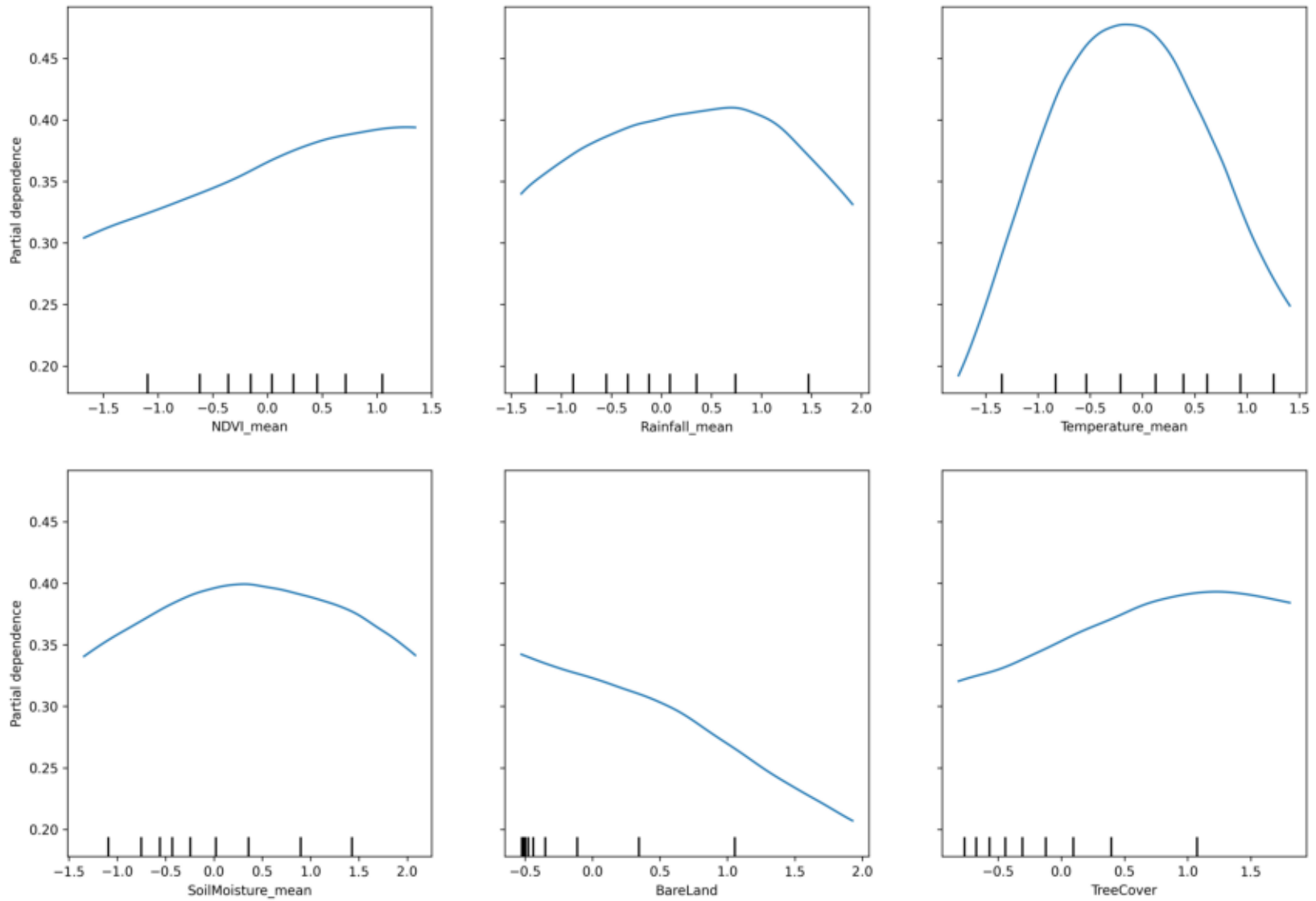
Roc Xgboost

ROC Curves (OvR): xgboost

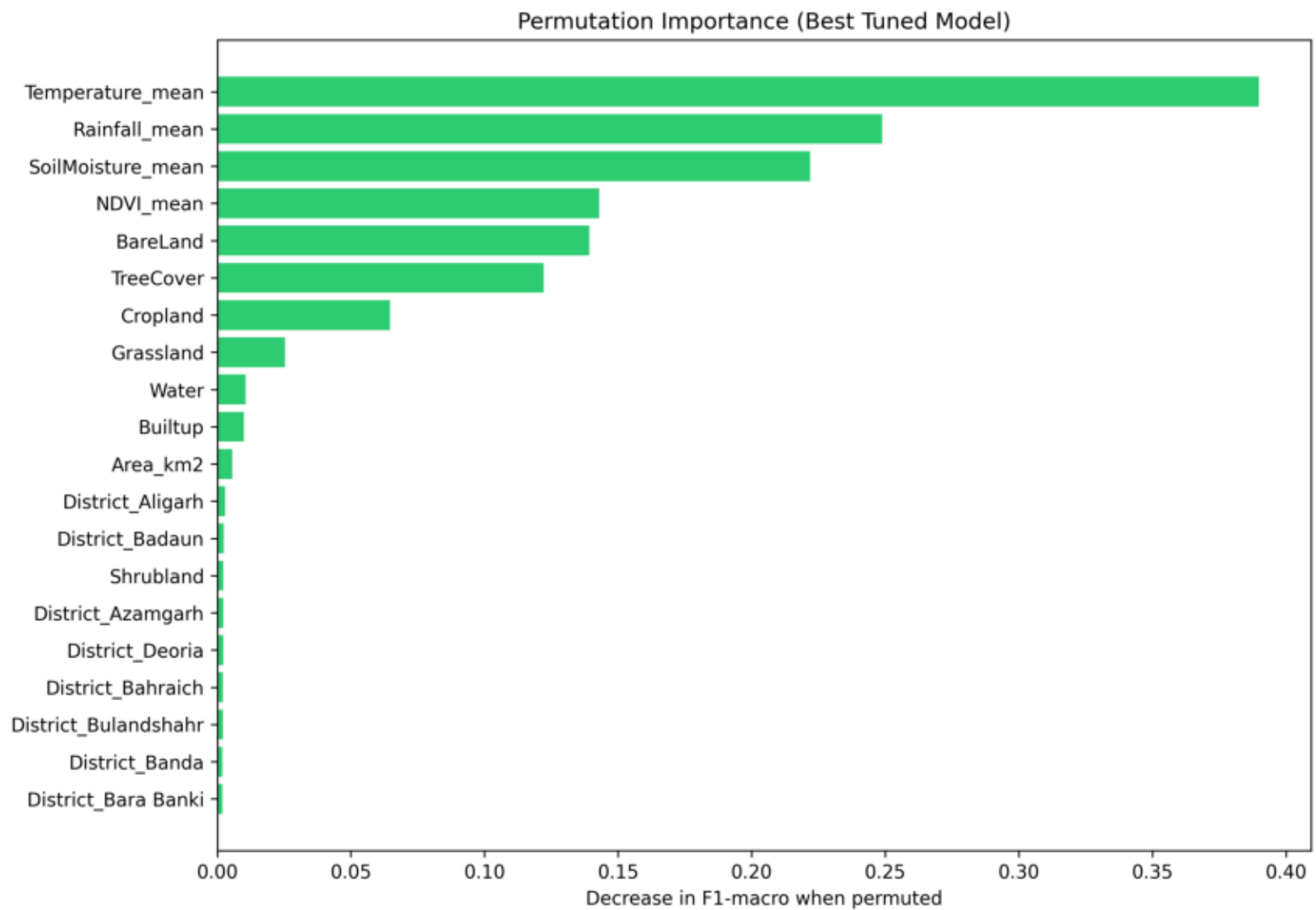


Partial Dependence

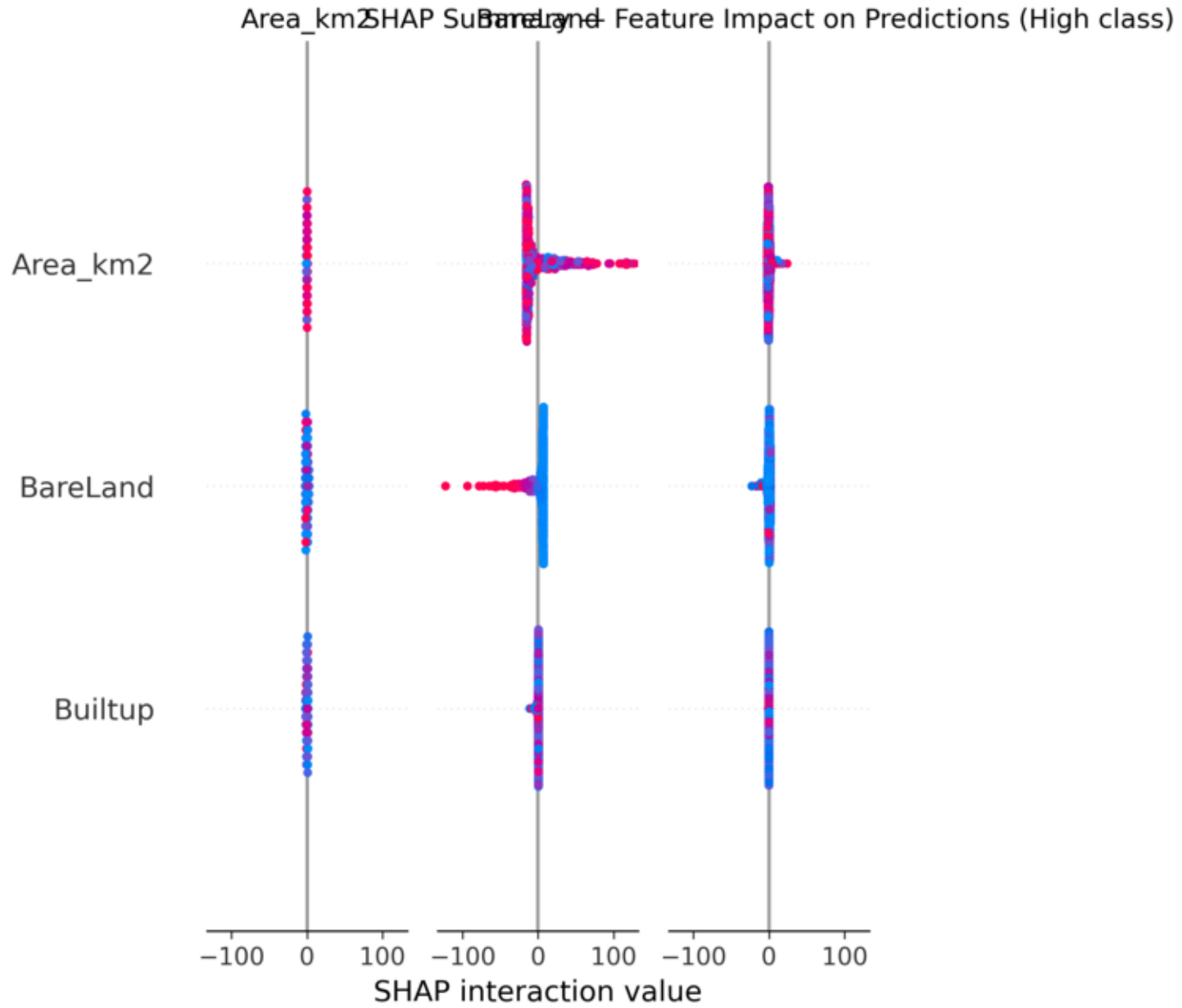
Partial Dependence Plots — Drivers of High Degradation Class

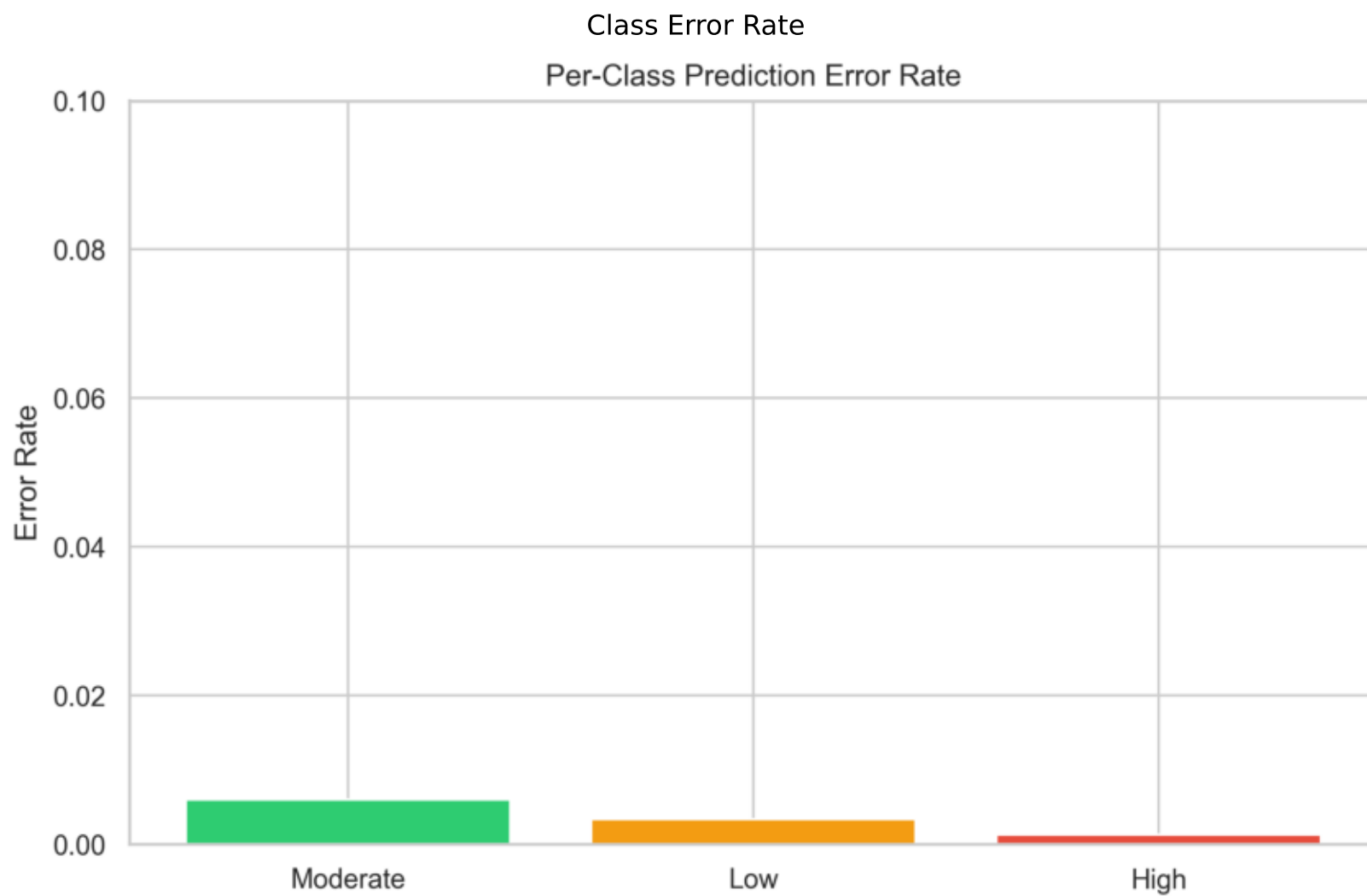


Permutation Importance

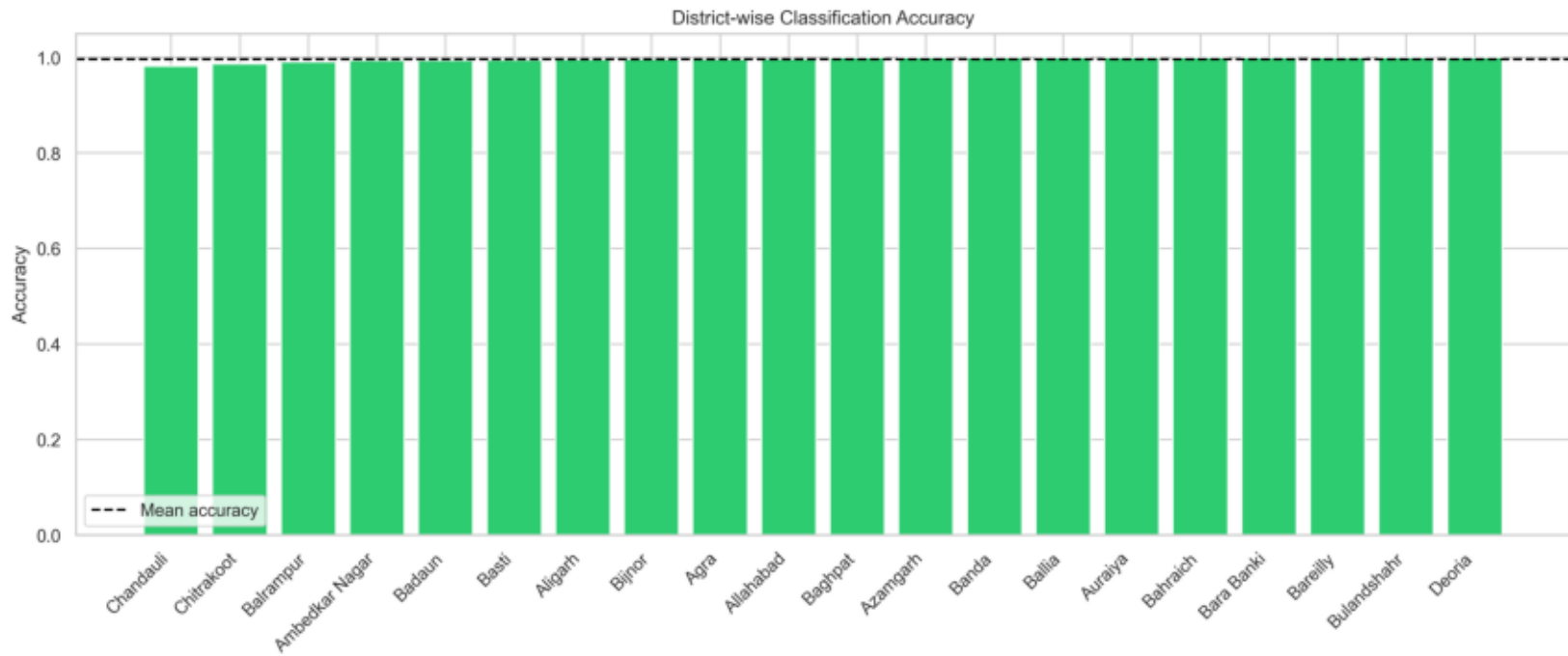


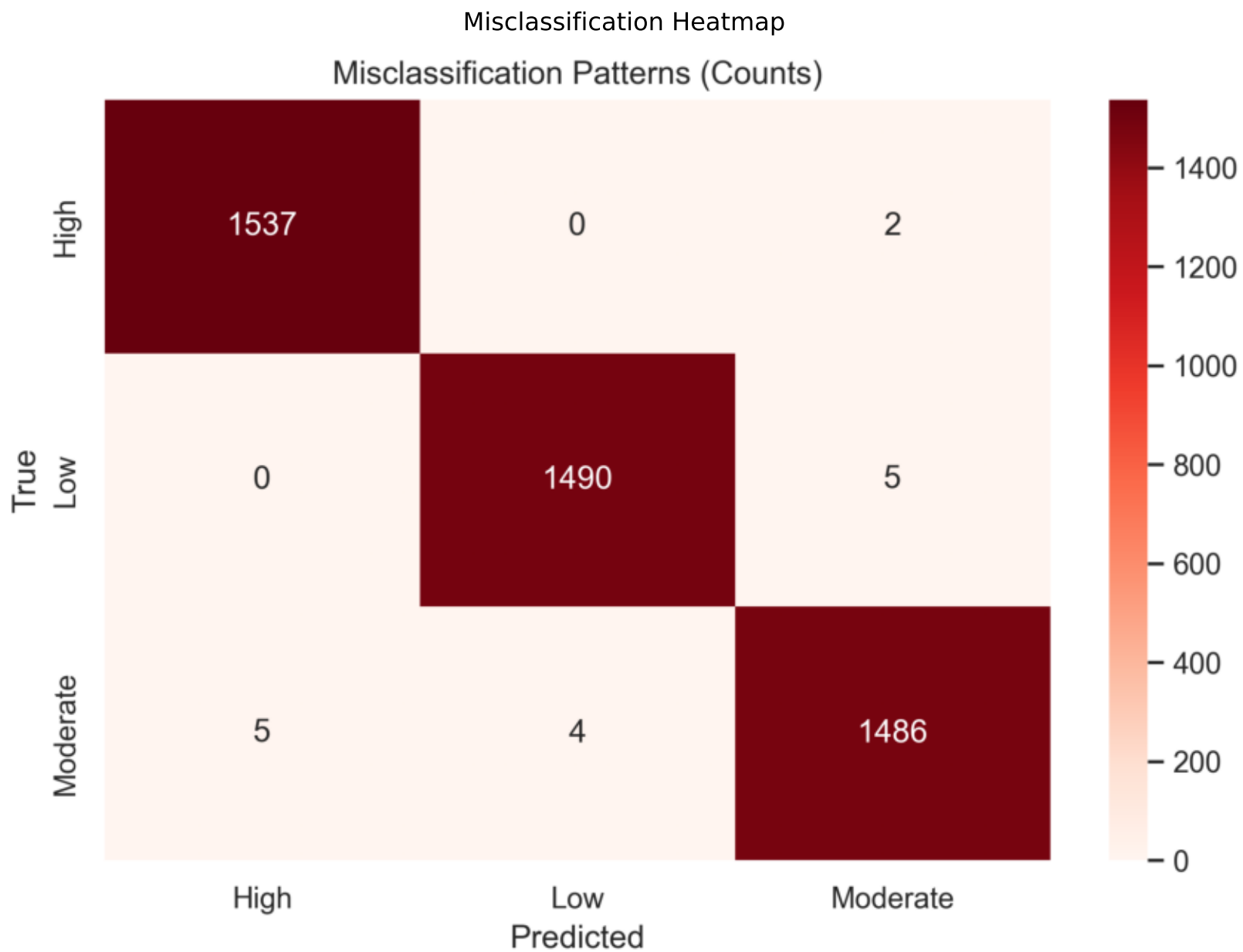
Shap Summary





District Accuracy





Year Accuracy
Year-wise Classification Accuracy

